

Identity-Consistent Analysis of Long-Shot Windsurfing Video

A Domain-Specific Offline Tracking System

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Abstract

Long-shot windsurfing video combines small targets, large camera pans, prolonged overlaps, and rapidly changing backgrounds. The desired output is not a generic MOT trace but a separate, stable rider-relative video for each surfer; one false identity merge can invalidate an otherwise useful result. We present an offline analysis system that detects surfers, forms conservative local tracklets, links them globally with camera-compensated motion and a foreground-masked sail-color descriptor, and uses two pose keypoints on the rig to drive a rider-relative virtual camera. The tracking stage is evaluated on 21 manually reconstructed development videos containing 41,004 retained observations. On this fixed-observation protocol, the production system achieves 0.957 pairwise precision, 0.918 recall, and 0.937 F1, compared with 0.792 F1 for OC-SORT and 0.828 for BoT-SORT. Compared with OC-SORT, it reduces fragmentation excess from 845 to 42, but nine of its 95 output tracks mix rider identities and these errors affect seven of the 21 videos.

Keywords: multi-object tracking; offline data association; camera motion compensation; sports video; windsurfing; pose-guided framing

1 Introduction

Shore-recorded windsurfing video is visually rich but difficult to repurpose into a focused view of one athlete. The camera follows the action with large pans, riders occupy few pixels, rigs overlap for extended periods, and an athlete may disappear behind another sail before returning against a different background. A useful system must do more than draw plausible boxes: it must preserve one physical identity long enough to produce a stable rider-relative video.

This requirement changes the engineering objective. A missed association leaves two fragments that can be reviewed or joined later. A false association silently inserts another surfer into the output and can invalidate the complete sequence. Standard online trackers are designed for causal decisions and are typically evaluated with metrics that balance detection and association errors. Here both failure modes matter but not equally: a false merge can corrupt a complete rider video, whereas excessive fragmentation can leave even pure tracks unusable. The full recording is available, so the system favors conservative local decisions and recovers continuity later through global association.

We describe the resulting end-to-end computer-vision system. A custom-trained pose detector supplies surfer boxes and two rig keypoints. Masked global-motion estimation separates camera pan from target motion. Conservative online association creates high-purity tracklets, after which an offline integer program links compatible tracklets using motion, temporal gap, and a sail-specific color representation. Finally, the pose keypoints define a temporally smoothed anchor and scale for a stable virtual camera. Source code, technical documentation, and an output demonstration are publicly available for inspection [12].

The contributions are:

- an explicit formulation of long-shot windsurfing analysis around asymmetric costs for identity merges and fragmentation, with complete-video processing;
- a domain-specific association pipeline combining foreground-masked sail appearance, camera-compensated four-dimensional motion, conservative tracklets, geometric vetoes, and global path optimization;
- a two-keypoint rider-relative stabilization signal designed around windsurf rig geometry rather than generic human pose; and
- a fixed-observation evaluation on 21 manually reconstructed videos, including OC-SORT and BoT-SORT comparisons, staged ablations, and a link-level failure audit.

2 Problem Regime and Requirements

The input is a complete windsurfing recording captured from shore. A scene usually contains only a handful of surfers, but each target can be small and several sails may remain close or overlap for seconds. The camera follows the action with pans of about 90° in demanding examples, sometimes more. Consequently almost every background feature moves in image coordinates, and newly revealed portions of the scene may share no features with earlier frames.

The desired result is one continuous, dense trajectory per physical surfer from which a focused video can be rendered. This gives three requirements.

Identity is the primary invariant. A track that switches rider once is not partly correct from the user’s perspective: the extracted sequence mixes identities. Lo-

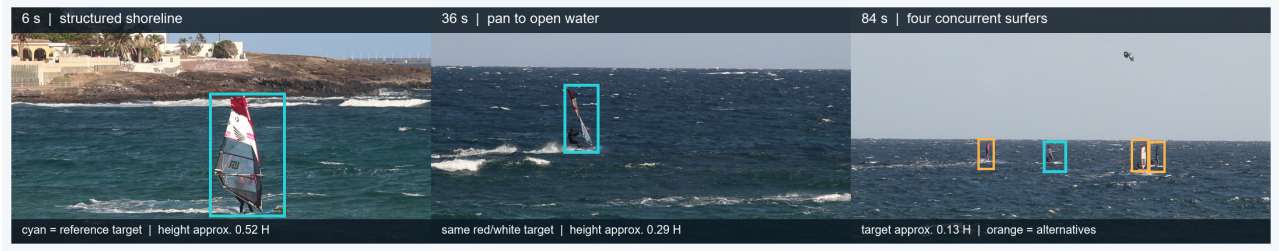


Figure 1: Operating regime in one recording at 6, 36, and 84 s. Cyan follows the same red-and-white sail while the background and camera direction change; the target height drops from approximately $0.52H$ to $0.13H$. Orange marks plausible competing surfers. The example combines scale change, camera motion, and identity ambiguity within one continuous shot.

calization smoothness and recall matter only after identity is preserved.

Association errors are asymmetric. A missed link creates fragments that remain inspectable. A false link combines two identities and is difficult to detect downstream. The system should therefore abstain when evidence is insufficient. “Near-zero false merges” is the design target, not a claim that the implementation achieves perfect tracking.

The complete recording is available. There is no real-time constraint. Causal processing is useful for forming local tracklets, but ambiguous choices need not be finalized before later evidence arrives. This motivates a two-stage design: conservative local association followed by global offline linking.

Large camera motion makes raw image displacement a poor association cue. The common background motion must first be estimated and applied to the state prediction. Long gaps remain intrinsically ambiguous: a later fragment may be a reappearance or a different surfer entering after the earlier track ended. Appearance is therefore required alongside motion, but the representation should describe the visible sail rather than a generic pedestrian.

3 Related Work

Online tracking by detection. SORT combines a Kalman filter with frame-wise assignment and established a strong, minimal online baseline [1]. Deep SORT adds a learned pedestrian appearance metric to survive longer occlusions [2], while ByteTrack improves continuity by associating low-confidence detections rather than discarding them immediately [3]. OC-SORT revisits motion estimation after occlusion and uses observations to correct accumulated prediction error [4]. BoT-SORT combines motion, camera compensation, and optional appearance features in a stronger pedestrian tracker [5]; Deep OC-SORT further adapts the contribution of appearance under feature degradation [6]. These systems motivate our motion and appearance cues, but remain fundamentally causal at association time.

Offline and global association. Batch tracking has long been formulated as network flow or related graph

optimization [7]. More recent work learns objectives for multi-frame flow association rather than training only local pair costs [8]. Our integer program belongs to this family but targets a small, bounded graph of already conservative tracklets. It is intentionally simple: start, end, discard, and link choices are optimized jointly, while the domain-specific work lies in constructing reliable fragments and costs.

Motion, appearance, and evaluation. Kalman filtering [10] and Rauch–Tung–Striebel smoothing [11] provide the temporal state machinery used before and after association. Learned ReID is effective when its training semantics match the target class, but pedestrian embeddings allocate capacity to body and clothing cues that occupy very few pixels in our footage. We instead exploit the sail: a large, saturated, deliberately distinctive surface. Standard MOT metrics such as HOTA explicitly balance detection, association, and localization [9]. Because our saved observations are fixed and the product loss is dominated by identity contamination, we report pair-clustering precision and recall together with absolute contaminated-track and fragmentation counts rather than claiming an end-to-end MOT score.

Pose-guided framing. The stabilization stage is related to pose-guided cropping, but its goal is not human pose recovery or action recognition. A windsurf rig is a large articulated object whose axis-aligned detector box changes strongly with rotation. Two semantic rig points are sufficient to construct a rider-relative anchor and scale, making the pose representation deliberately task-specific.

4 System Overview

The pipeline separates perception, identity, and presentation geometry:

$$\begin{aligned}
 \text{video} &\rightarrow \text{boxes} + \text{pose keypoints} \\
 &\rightarrow \text{local tracklets} \\
 &\rightarrow \text{offline identity paths} \\
 &\rightarrow \text{stable rider views.}
 \end{aligned} \tag{1}$$

The detector is a custom-trained one-class YOLO11m-pose model. It was initialized from public pretrained

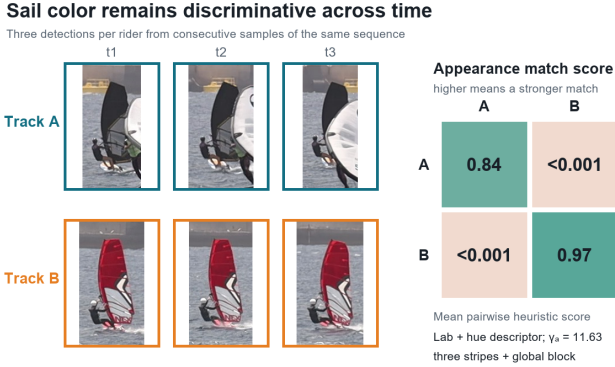


Figure 2: Production color descriptor on consecutive crops of two sails. Repeated observations score highly, while cross-sail matches reach the lower bound. The hand-selected example explains the cue rather than measuring it.

weights and trained for 700 epochs at 640-pixel input resolution on manually reviewed boxes and two windsurf-specific pose keypoints: the boom–mast junction and mast tip. Each box is also converted into a compact sail-color descriptor after border-connected background suppression. Independently, background features are estimated between adjacent frames while masking detected surfers; the resulting camera transforms are passed to both tracking stages.

The first association stage is causal but deliberately conservative. It accepts only locally unambiguous matches and exposes uncertainty as separate tracklets. The second stage sees the complete recording, creates a bounded graph of feasible tracklet continuations, and solves for disjoint paths. Post-processing smooths linked boxes and interpolates internal gaps, but retains the distinction between observed and synthesized detections. The pose stage then converts every dense identity track into a frame-indexed virtual-camera signal.

The components have explicit contracts. Detection proposes observations; local association should prioritize fragment purity; global association should prefer a split over implausible identity evidence; and stabilization must never change track identity. This modularity is important because only the association contract is evaluated against alternative trackers, while the final view also depends on detector and pose-keypoint quality.

5 Offline Identity Association

The tracker converts detections into high-purity local fragments and then into globally consistent paths. The same camera-motion estimates support both stages.

5.1 Foreground-masked sail appearance

Windsurfing sails are large, saturated, and often visually distinctive even when the rider occupies only a few pixels. For each detection crop we build a descriptor from a joint histogram of the chromatic Lab channels (a, b), a circular hue histogram, and a lower-weight lu-

minance histogram. Saturation weighting emphasizes colored sail panels over gray water, spray, and haze. Three vertical panels preserve coarse color layout, and a fourth block represents the complete crop. Each block is L1-normalized and square-root mapped before the concatenated vector is L2-normalized, yielding a bounded Hellinger-style distance.

Figure 2 illustrates the descriptor response on two visually distinct sails.

The descriptor first removes likely background using a border-derived Lab model. A gradient gate identifies low-texture pixels consistent with that model, and only candidate background components connected to the crop border are removed. After morphological cleanup, only the largest remaining foreground component is retained; a minimum-mask fallback avoids empty descriptors. This foreground mask is important: an earlier inactive code path used a fixed center rectangle, allowing water and overlapping sails to dilute the intended cue.

Figure 3 contrasts the locally ambiguous online view with the later fragment evidence available to the offline optimizer.

For a tracklet, all observation embeddings are averaged into a whole-fragment prototype. This replaced endpoint-biased exponential averages, which emphasized the last crop of the earlier fragment and first crop of the later fragment—exactly the frames most likely to contain both sails during a crossing. Given descriptor distance d , the current heuristic match score is

$$\epsilon = 10^{-6},$$

$$S_a = \text{clip}_{[\epsilon, 1-\epsilon]}(1 - \gamma_a d), \quad \gamma_a = 11.63. \quad (2)$$

It is a tuned similarity score, not a calibrated posterior probability; the graph uses $C^{\text{appearance}} = -\log S_a$. The coefficient γ_a was tuned on the development set. The numerical margin ϵ keeps the logarithm well defined.

5.2 Conservative local tracklets

Each detection initially forms a singleton. A greedy preprocessor advances a constant-velocity Kalman state [10] through the estimated frame-to-frame camera transforms and compares the prediction with current detections. Strict links require both motion and appearance evidence; looser links are accepted only when they are unique. Duplicate proposals, competing successors, or multiple plausible candidates remain separate. The local stage therefore sacrifices completeness to preserve fragment purity.

This behavior is deliberate. In the evaluation reconstruction, 598 of 601 preprocessor fragments are pure with respect to retained gold identities. The dominant catastrophic error is introduced later when two pure fragments are linked incorrectly, which makes the offline edge decision the appropriate place for additional safeguards.

5.3 Candidate graph

For non-overlapping fragments T_i and T_j , a directed candidate $i \rightarrow j$ is created only when the gap is at most

Offline global association resolves ambiguous gaps

A causal tracker must choose before later fragments arrive; offline linking can use them.

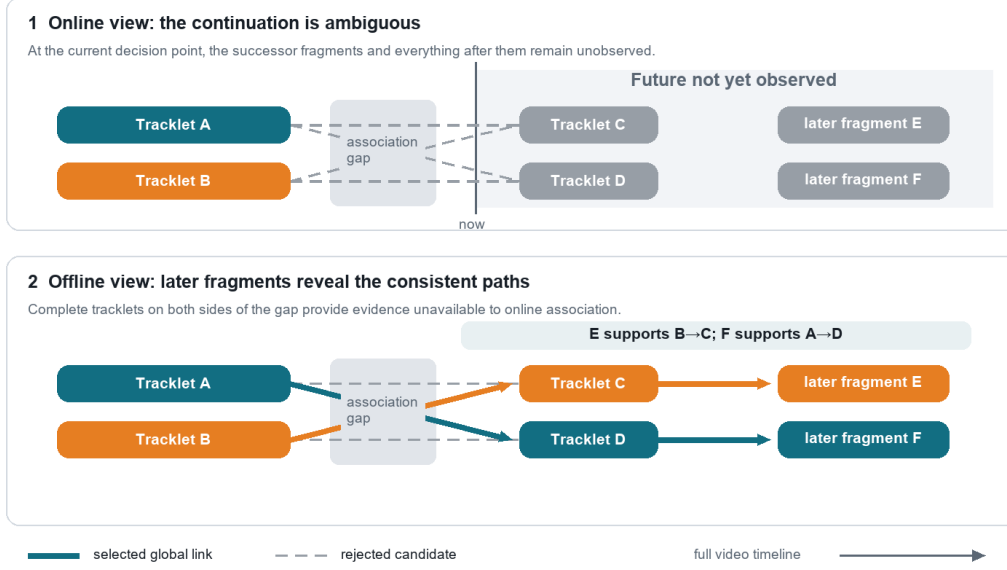


Figure 3: Future fragment evidence supports global tracklet association. Later stable observations distinguish locally ambiguous successors; the optimizer then chooses a compatible pairing jointly. Start, end, discard, and gap costs are omitted from the conceptual diagram.

three seconds. A Kalman state fitted to all detections of T_i is propagated to the first observations of T_j , applying global camera transforms at every intervening frame. Unlike the earlier production call, the gate uses the complete measurement (c_x, c_y, w, h) rather than discarding width and height.

Every candidate receives

$$\begin{aligned} C_{ij} &= w_m C_{ij}^{\text{motion}} + w_a C_{ij}^{\text{appearance}} \\ &\quad + w_g C_{ij}^{\text{gap}}, \\ C_{ij}^{\text{gap}} &= -\Delta f_{ij} \log p_{\text{miss}}. \end{aligned} \quad (3)$$

For each of the first two observations of T_j , the predicted Kalman state gives squared Mahalanobis distance d^2 over (c_x, c_y, w, h) . Motion cost is the mean negative log chi-square survival score, $-\log \Pr(\chi_4^2 \geq d^2)$; appearance uses the whole-fragment prototypes; and the gap term increases with the number of missing frames.

The graph now applies absolute plausibility vetoes before optimization. Edges are removed when motion cost exceeds 8, or when endpoint box area or aspect ratio changes by more than a factor of three. The Kalman filter already models size; the explicit ratios supplement it with an interpretable discontinuity bound. These rules implement abstention at the evidence boundary. They are necessary because a central fragment’s start and end penalties can otherwise make even poor finite evidence cheaper than leaving two trajectories separate.

5.4 Global path optimization

Binary variables represent starting or ending a path at each fragment, selecting a candidate edge, or discarding an eligible short fragment. Each retained fragment has exactly one incoming choice and one outgoing choice, so

selected links form disjoint, time-ordered paths. Start and end costs are reduced near image borders, where a surfer entering or leaving the field of view is more plausible. An edge that costs more than ending T_i and starting T_j independently is dominated and removed; the remaining outgoing degree is bounded before solving the integer program.

The method is offline in a precise sense: it compares stable fragments on both sides of a gap and chooses all compatible continuations jointly. It does not infer through arbitrarily long absence. Gaps beyond the horizon remain fragmented, and ambiguous evidence should do the same.

5.5 Dense trajectories

Linked detections remain sparse. The post-processor removes tracks that are too short or sparsely observed, performs a forward camera-compensated Kalman pass and backward Rauch–Tung–Striebel smoothing [11], and interpolates internal gaps. The dense trajectory records whether each observation was detected or synthesized, then feeds the rider-relative stabilization stage.

6 Pose-Guided Stabilization

Tracking determines *which* surfer is present; the downstream view must decide where to place a virtual camera and how tightly to crop. An axis-aligned detector box is a poor control signal because a tall, rotating rig changes box center and height even when the athlete moves little. Direct box-based cropping therefore produces lateral drift and visible zoom pumping.

Bounding-box and pose-guided framing geometry



Figure 4: Pose-derived virtual-camera geometry. The detector box and center (orange) describe image extent. Boom junction and mast tip (cyan) provide a rider-relative anchor (magenta) and scale cue.

6.1 Two task-specific pose keypoints

The pose model predicts the boom–mast junction and mast tip. These keypoints were chosen for rendering geometry rather than as a reduced human skeleton. The boom junction provides a repeatable reference near the rider; the boom-to-tip distance provides a rig-relative scale cue that is less sensitive to in-plane rotation than box height. The checked-in development dataset contains 1,123 manually labeled training images and 56 validation images. The split is image-level and can contain frames from the same source sequence on both sides, so it does not establish cross-sequence generalization.

Figure 4 summarizes the detector geometry, the two predicted keypoints, and the derived anchor and scale signals used by the virtual camera. The figure makes the distinction visible: the box center follows the axis-aligned extent of the complete rig, whereas the pose-derived anchor remains tied to the boom–mast region near the rider. As the sail rotates, the box center can move even when the rider moves little; using the pose-derived point therefore reduces this geometric source of lateral drift before temporal smoothing.

For box top y_1 , height h , and center (c_x, c_y) , the fallback anchor is

$$\mathbf{a}_{\text{box}} = (c_x, y_1 + 0.6h). \quad (4)$$

When the boom junction $\mathbf{b} = (b_x, b_y)$ is visible, a low-noise mast proxy $\tilde{\mathbf{t}} = (b_x, y_1)$ gives

$$\mathbf{a}_{\text{pose}} = 0.15\tilde{\mathbf{t}} + 0.85\mathbf{b}. \quad (5)$$

The measured mast tip is intentionally excluded from position: its long lever arm amplifies localization noise. It remains useful for scale.

Small crops do not support equally reliable keypoints. The unsmoothed anchor blends pose and box geometry according to the shorter box dimension, with pose weight rising from zero at 45 pixels to one at 55 pixels. Missing boom observations are interpolated for short gaps; longer gaps ease toward the per-frame box fallback and back to pose when reliable evidence returns. A five-frame robust weighted mean suppresses isolated jumps while preserving response to actual motion.

6.2 Rig-relative scale

When both keypoints are visible, scale begins with $\ell = \|\mathbf{b} - \mathbf{t}\|_2$. Short gaps are interpolated or held; otherwise the anchor-to-box-top distance supplies a fallback. A centered 31-frame mean (radius 15) produces $\bar{\ell}$, and the requested crop height is

$$C = \max\left(\frac{\bar{\ell}}{0.75}, \frac{h}{0.9}, 1\right), \quad S_m = \text{clip}_{[\epsilon, 1]}\left(\frac{C}{H}\right), \quad (6)$$

where H is source height and S_m is the mast-relative crop scale. The mast segment is targeted to 75% of the output, while the box term prevents excessively tight crops. The resulting anchor and scale are stored per frame with the identity track. Global background stabilization remains a separate signal: it improves motion estimation and optional overview rendering, but cannot replace rider-relative composition when the rig rotates within the scene.

7 Evaluation

We evaluate identity association on 21 manually curated development videos. The benchmark fixes the universe of 41,079 saved, non-interpolated detections and gives every method the same bounding boxes. Of these, 41,004 belong to retained surfer identities and 75 are annotated for rejection. This deliberately isolates association from detector recall: the results answer whether the available observations were grouped correctly, not whether YOLO found every surfer.

The saved observations were selected during manual trajectory reconstruction and their boxes were Rauch–Tung–Striebel smoothed within gold identity trajectories. This is stronger leakage than merely conditioning on useful boxes: the motion cue receives geometry derived from the target identity partition. The same videos also informed system design, and the provenance of the original ILP parameter tuning is not fully recorded. The benchmark is therefore an in-sample engineering diagnostic and failure analysis, not a clean end-to-end or held-out tracker comparison.

7.1 Metrics

For retained detections, pairwise precision and recall compare every unordered pair. A pair is positive when both observations have the same annotated identity:

$$P_{\text{pair}} = \frac{\text{correct same-identity pairs}}{\text{all pairs joined by the tracker}}, \quad (7)$$

$$R_{\text{pair}} = \frac{\text{correct same-identity pairs}}{\text{all annotated same-identity pairs}}. \quad (8)$$

False merges reduce precision and fragmentation reduces recall. Missing outputs are treated as singleton predictions rather than being silently omitted. Micro scores pool all pairs and consequently weight long identities quadratically; macro scores average per-video values. Coverage is the fraction of retained observations emitted by a method. A contaminated predicted track contains more than one gold identity; “contaminated observations” counts every observation in such a track and therefore does not distinguish a brief intrusion from a wholly wrong trajectory. If k_g is the number of predicted pieces containing observations of gold identity g , then fragmentation excess is $\sum_g \max(k_g - 1, 0)$, including pieces shared through a contaminated prediction. An exact video has full coverage and a partition identical to gold up to track-label permutation.

7.2 Fixed-observation comparison

Table 1 compares the production pipeline with OC-SORT and BoT-SORT. OC-SORT receives the fixed detections directly. BoT-SORT uses its ECC camera compensation but no pedestrian-specific ReID network. The production method reruns conservative preprocessing, masked video stabilization, and offline ILP association. Its sail descriptor uses border-derived foreground masking and a whole-fragment appearance prototype; motion gating evaluates center, width, and height; and implausible motion, area, or aspect-ratio transitions are excluded before optimization.

Production reduces fragmentation from 845 to 42 relative to OC-SORT while retaining the same absolute number of contaminated tracks. This continuity gain comes with lower pair precision, a larger contaminated-output fraction (9/95 versus 9/165), and more observations contained in contaminated tracks (9,982 versus 6,184). Eleven of 21 videos are exact, compared with four for BoT-SORT and one for OC-SORT. The result is the selected operating point for this application, but not a uniform improvement and not the nominal zero-false-merge target. OC-SORT’s lower mixed-track rate comes with 165 outputs for 88 identities, 845 excess fragments, and 1.9% of retained observations left unassigned.

A 10,000-sample bootstrap that resamples complete videos gives 95% intervals for micro pair F1 of [0.721, 0.864] for OC-SORT, [0.759, 0.909] for BoT-SORT, and [0.852, 0.995] for production. The broad, overlapping intervals reinforce the development-set interpretation rather than a general ranking claim. The resampling procedure and seed are recorded with the evaluation artifacts [12].

7.3 Safeguard ablation

The initial production reconstruction exposed a concentrated false-link problem. Preprocessor fragments were

almost always identity-pure, but the ILP sometimes joined two pure fragments from different surfers. Many of the unexpected short-gap errors occurred at crossings, where the endpoint crops contained both sails. The previous appearance model emphasized precisely those boundary frames, the color descriptor’s intended foreground mask was bypassed by an early return, and the ILP requested a position-only projection from a Kalman model that already represented width and height.

Table 2 records two staged revisions. Because several changes are bundled in the first stage, this is a system ablation rather than a claim that any single component produced the full difference. Foreground masking, whole-fragment appearance aggregation, and four-dimensional motion evaluation raise micro F1 from 0.893 to 0.928. Adding conservative vetoes for motion cost and threefold area or aspect-ratio discontinuities raises precision further and halves contaminated tracks, without increasing fragmentation.

The vetoes are not a second tracker layered on top of the Kalman filter. The Kalman state already contains center, width, and height; the earlier ILP call discarded the size dimensions. The revised system uses the full four-dimensional Mahalanobis evidence and additionally refuses links whose absolute evidence is implausible. This second step is necessary because the ILP’s start and end penalties can otherwise make even a very poor link cheaper than splitting two central fragments.

A manual audit found nine wrong selected fragment joins across six videos. This audit counts only joins between two individually pure preprocessor fragments. Contaminated production outputs span seven videos because evaluation sequence 15 instead contains an already impure local fragment (225 observations from one identity and one from another), so its adjacent joins are excluded from the pure-to-pure edge count. Unlike the removed geometric failures, they concentrate at crossings, visually near-identical sails, and crops containing both competitors. Their implications and the unsuccessful global-margin abstention experiment are discussed in Section 8.

8 Discussion

8.1 What the comparison establishes

The benchmark quantifies the intended trade-off. OC-SORT has the highest pair precision and nine identity-mixed outputs, but 845 excess fragments make its result operationally unsuitable. Production also has nine identity-mixed outputs—a larger fraction of its 95 outputs—while reducing fragmentation to 42. Aggregate F1 is therefore a diagnostic rather than an acceptance criterion; the product still requires complete-trajectory review.

8.2 Residual failure mode

The original failure audit separated two kinds of false link. Geometrically implausible joins resulted from by-

Table 1: Fixed-observation identity-association results on 21 development videos. “Contam.” is contaminated tracks over emitted tracks; “C. obs.” counts all observations contained in those tracks.

Method	Coverage	Pair P	Pair R	Pair F1	Macro F1	Contam.	C. obs.	Frag. excess
OC-SORT	0.981	0.985	0.662	0.792	0.857	9/165 (5.5%)	6,184	845
BoT-SORT, no ReID	0.996	0.920	0.752	0.828	0.858	16/142 (11.3%)	14,459	244
Production offline ILP	0.999	0.957	0.918	0.937	0.970	9/95 (9.5%)	9,982	42

Table 2: Staged production-pipeline revision on the same development set.

Configuration	Pair P	Pair R	Pair F1	Contaminated tracks	Fragmentation excess
Original reconstruction	0.907	0.880	0.893	20	48
+ foreground mask, fragment mean, 4D motion	0.939	0.917	0.928	17	44
+ conservative link vetoes	0.957	0.918	0.937	9	42

passed foreground masking, endpoint-biased appearance aggregation, and position-only use of a size-aware motion model. The revised evidence and conservative vetoes remove many of these cases. The remaining wrong joins cluster at crossings, visually near-identical sails, and crops containing both competitors. In those cases, time, position, scale, and color can all support the wrong continuation.

This residual mode is not well addressed by another global scalar threshold. An abstention experiment removed each chosen edge in turn and compared the alternative objective. The resulting margin did not separate wrong from correct links: useful thresholds rejected substantially more correct associations than errors. A credible next step would instead model the interaction explicitly—for example, by detecting paired crossings and comparing pre- and post-overlap appearance jointly. We leave that extension unimplemented rather than tuning a development-set-specific margin.

8.3 Scope and limitations

The videos are a development set, and identity-conditioned box smoothing leaks information into motion; this is neither an end-to-end detector test nor a clean MOTA or HOTA comparison. BoT-SORT omits its pedestrian ReID model, while pose-guided framing has no independent cross-video stability benchmark. Its image-level split may also contain nearby frames from the same source video.

The three-second horizon limits false long-range joins and graph size, but longer genuine disappearances remain split. The system therefore combines conservative automation with review rather than guaranteeing perfect recovery.

9 Conclusion

We presented an offline system that turns long-shot windsurfing video into stable, rider-relative trajectories. Conservative local tracklets, camera-compensated four-dimensional motion, sail-specific color, global path optimization, and two rig pose keypoints implement a common design: preserve identity under ambiguity and use future fragment evidence before committing to longer

associations.

On the conditional development-set protocol, production reduces fragmentation from 845 to 42 relative to OC-SORT while retaining nine identity-mixed tracks. The staged safeguards reduce identity-mixed tracks from 20 to nine. This is not a general tracker ranking; it shows how domain structure, offline inference, and an asymmetric failure cost can turn custom pose detections into useful rider-relative views.

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A Reconstruction Details

A.1 Evaluation reconstruction

The benchmark uses 21 manually reconstructed development videos. It contains 41,079 non-interpolated saved observations: 41,004 assigned to 88 retained identities and 75 assigned to the discard class. Saved boxes were RTS-smoothed after manual identity reconstruction. For the production method, the historical intermediate tracklets were unavailable; the evaluator therefore starts from singleton saved observations, reruns the current preprocessor, and then runs the current ILP tracker. A missing method output is scored as a unique singleton rather than omitted from the pair metrics.

Predictions were generated with implementation commit `a76c2bd774f739e40e1f15a80574ca99bb34d236`. The final repository snapshot, aggregate JSON, per-video CSV, protocol notes, and bootstrap script are published as the project and evaluation artifacts [12]. The annotated videos and gold reconstruction files are not distributed, so these files form an auditable result snapshot rather than an independently rerunnable artifact. BoxMOT distribution 13.0.17 was used; its package-level version string reports 13.0.16.

Runtime is retained only as an operational diagnostic. On the evaluation machine, OC-SORT took 9.0 seconds, BoT-SORT 497.7 seconds, and the production association path 402.3 seconds. These are not comparable speed measurements: OC-SORT needs no video decode, BoT-SORT includes decode and ECC, and production includes masked stabilization but excludes initial crop-embedding extraction.

A.2 Production configuration

The reported production run uses a maximum candidate gap of three seconds and at most ten outgoing candidate links per fragment. The ILP start and end costs are both 96.0993. Motion, appearance, and gap costs have weights 3.4694, 2.2492, and 9.9926; the per-frame miss probability is 0.8037 and the appearance coefficient γ_a is 11.6301. The implementation uses the numerical probability margin $\epsilon = 10^{-6}$ for appearance-score clipping. Motion compares the first two detections of a successor using the complete (c_x, c_y, w, h) measurement. Candidate links are rejected when the motion cost exceeds 8 or when the area or aspect-ratio change factor exceeds 3.

The appearance prototype retains the full fragment. Tracklets with at most nine detections may be dis-

carded, with first-detection cost 36.4497 and growth factor 1.6423. These values are reported for reconstruction, not as independently validated universal hyperparameters.

For normalized fragment-end center (x, y) and border margin $m = 0.1$, define $f = \min(1, \max(f_x, f_y))$, where $f_x = \max(0, m - x, x - (1 - m))/m$ and likewise for f_y . Start and end costs are $w(1 - f/2)$: full cost in the central 80% of the image and half cost at the border. Sensitivity to the three-second horizon and the motion, area, and aspect-ratio veto thresholds was not measured.

A.3 Baseline configuration

OC-SORT uses the frozen BoxMOT defaults recorded by the evaluator: confidence thresholds 0.1 and 0.2, maximum age 30, minimum hits 3, IoU threshold 0.3, $\Delta t = 3$, inertia 0.2, BYTE association disabled, $Q_{xy} = 0.01$, and $Q_s = 0.0001$. BoT-SORT uses ECC global-motion compensation with ReID disabled; its high, low, and new-track thresholds are 0.5, 0.1, and 0.6, its buffer is 30, its match threshold is 0.8, and score fusion is disabled. All methods receive the same fixed observations.

A.4 Detector checkpoint

The supporting detector is YOLO11m-pose initialized from `yolo11m-pose.pt` and trained for 700 epochs at 640×640 input resolution with deterministic seed 0. The released checkpoint is `video_processing/inference/weights/yolo_models/windsurfing_pose/best.pt`, with SHA-256 `4c475caeba9ede3cfc56277161284a4a703be9ff46ae8735a788aac093ca9369`. The image-level development split contains 1,123 training and 56 validation images.

A.5 Audited wrong links

The final link audit found nine wrong selected links in six videos: one each in evaluation sequences 12, 16, and 18, and two each in sequences 19–21. These neutral indices follow the sequence order in the checked-in per-video result file. This count concerns selected fragment edges and is distinct from the number of contaminated output tracks or observations. In particular, the nine contaminated outputs span seven videos: the edge audit counts only wrong links between pure fragments. The seventh sequence, evaluation sequence 15, instead contains one impure local fragment with a 225-to-1 identity mixture and is therefore excluded from that classification.